

Unit 6.3 The dot product

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Definition: If $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$ are vectors in $\mathbb{R}^2$, then the dot product of $\vec{a}$ and $\vec{b}$ is the number (scalar) where

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2$$

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, are vectors in $\mathbb{R}^3$, then

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Examples: 1. $\langle -2, 3 \rangle \cdot \langle 4, -5 \rangle \stackrel{\text{Sol.}}{=} -2(4) + 3(-5) = -23$

2. $(8\vec{i} - 2\vec{j} + 3\vec{k}) \cdot (2\vec{i} + 4\vec{k}) \stackrel{\text{Sol.}}{=} \langle 8, -2, 3 \rangle \cdot \langle 2, 0, 4 \rangle$
 $= 8(2) + (-2)(0) + 3(4)$
 $= 16 + 12$
 $= 28$

3. $\langle 2, 3, -1 \rangle \cdot \vec{k} \stackrel{\text{Sol.}}{=} \langle 2, 3, -1 \rangle \cdot \langle 0, 0, 1 \rangle$
 $= 2(0) + 3(0) + (-1)(1)$
 $= -1$

Properties of the dot product:

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be vectors in $\mathbb{R}^2$ (or $\mathbb{R}^3$) and $k \in \mathbb{R}$ (k scalar)

1) $\vec{a} \cdot \vec{a} = |\vec{a}|^2$

2) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

$$3) \quad \bar{a} \cdot (\bar{b} + \bar{c}) = \bar{a} \cdot \bar{b} + \bar{c} \cdot \bar{a}$$

$$4) \quad (k\bar{a}) \cdot \bar{b} = k(\bar{a} \cdot \bar{b}) \quad \text{and}$$

$$5) \quad \bar{0} \cdot \bar{a} = \bar{0}$$

Proof 1 let $\bar{a} = \langle a_1, a_2 \rangle$ be a vector in $\mathbb{R}^2$

$$\text{Then L.H.S} \quad \bar{a} \cdot \bar{a} = \langle a_1, a_2 \rangle \cdot \langle a_1, a_2 \rangle$$

$$= a_1 a_1 + a_2 a_2$$

$$= a_1^2 + a_2^2$$

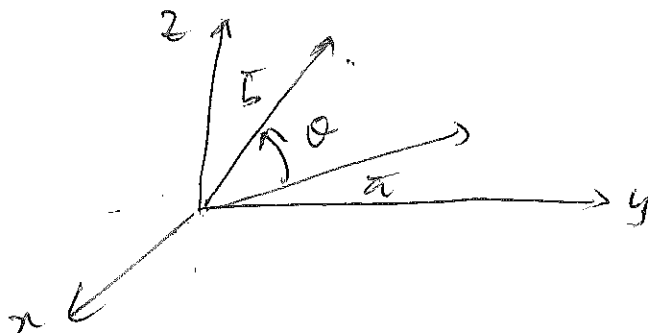
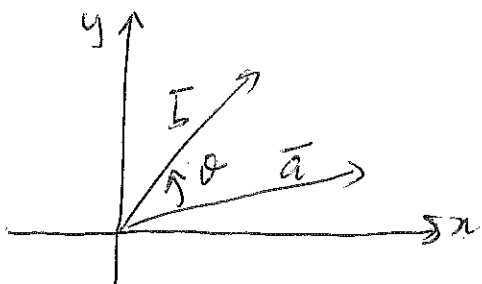
$$\text{R.H.S} \quad |\bar{a}|^2 = (\sqrt{a_1^2 + a_2^2})^2$$

$$= a_1^2 + a_2^2$$

$$\Rightarrow \text{L.H.S} = \text{R.H.S}$$

Geometric Interpretation

The dot product $\bar{a} \cdot \bar{b}$ can be interpreted in terms of the angle θ ($0 \leq \theta \leq \pi$) between $\bar{a}$ and $\bar{b}$ ($\bar{a}$ and $\bar{b}$ start at the origin).



If $\vec{a}$ and $\vec{b}$ are parallel vectors,
then $\theta = 0$ or $\theta = \pi$

Theorem

If θ is the angle between the vectors
 $\vec{a}$ and $\vec{b}$, then

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

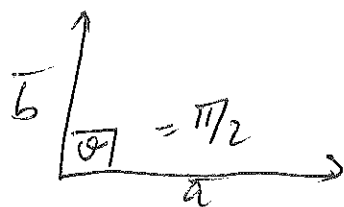
Corollary

If $\vec{a}$ and $\vec{b}$ are non-zero vectors
then (we can deduce from the Theorem)

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Definition

Two vectors $\vec{a}$ and $\vec{b}$ are called
orthogonal (perpendicular) if the angle
between the vectors is $\theta = \frac{\pi}{2}$



Definition : Two non-zero vectors $\vec{a}$ and
 $\vec{b}$ are orthogonal, if and only if
 $\vec{a} \cdot \vec{b} = 0$

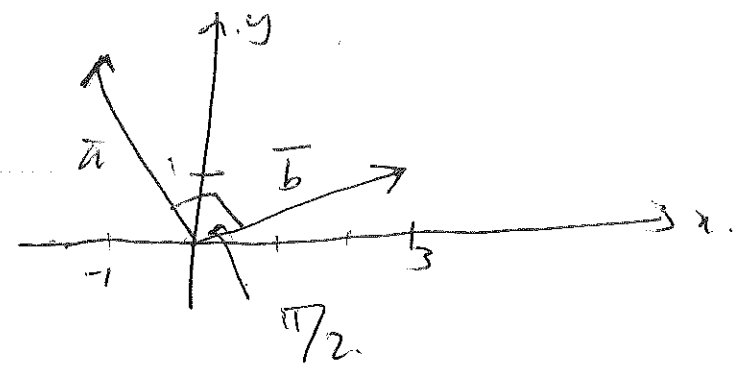
Example:

If $\vec{a} = \langle -1, 3 \rangle$ $\vec{b} = \langle 3, 1 \rangle$

then $\vec{a} \cdot \vec{b} = (-1)(3) + (3)(1)$
 $= -3 + 3$
 $= 0$

$\Rightarrow \vec{a}$ is orthogonal to $\vec{b}$

Graphically



Example: Find k such that

$\vec{a} = \langle -1, 3, k \rangle$ and

$\vec{b} = \langle 2, k, 4 \rangle$ are orthogonal

Solution: $\vec{a}$ and $\vec{b}$ will be orthogonal
if $\vec{a} \cdot \vec{b} = 0$

$\Rightarrow \langle -1, 3, k \rangle \cdot \langle 2, k, 4 \rangle = 0$

$\Rightarrow -2 + 3k + 4k = 0$

$\Rightarrow k = 2/7$

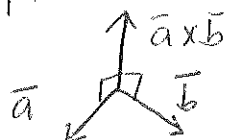
Unit 6.4

The cross product of vectors in $\mathbb{R}^3$

Definition. If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ then the cross product of $\vec{a}$ and $\vec{b}$ is the VECTOR defined by

$$(*) \quad \vec{a} \times \vec{b} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle (*)$$

Cross product is only defined when $\vec{a}$ and $\vec{b}$ are in $\mathbb{R}^3$



To understand how we calculate the cross product we first introduce the determinants.

Determinant of order 2:

$$\bullet \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Determinant of order 3

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

$$\begin{aligned} \text{eg } \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} &= \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ &= \vec{i}(a_2 b_3 - a_3 b_2) - \vec{j}(a_1 b_3 - a_3 b_1) + \vec{k}(a_1 b_2 - a_2 b_1) \end{aligned}$$



Thus the cross product of two vectors in $\mathbb{R}^3$ is

(2)

$$\vec{a} \times \vec{b} = \langle a_1, a_2, a_3 \rangle \times \langle b_1, b_2, b_3 \rangle$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$= \vec{i}(a_2 b_3 - a_3 b_2) - \vec{j}(a_1 b_3 - a_3 b_1) + \vec{k}(a_1 b_2 - a_2 b_1)$$

$$= \vec{i}(a_2 b_3 - a_3 b_2) + \vec{j}(a_3 b_1 - a_1 b_3) + \vec{k}(a_1 b_2 - a_2 b_1) \quad (*)$$

Example: Find $\vec{a} \times \vec{b}$ if $\vec{a} = \langle 1, 2, 0 \rangle$ and $\vec{b} = \langle 3, -1, 2 \rangle$

Solution: $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 0 \\ 3 & -1 & 2 \end{vmatrix}$

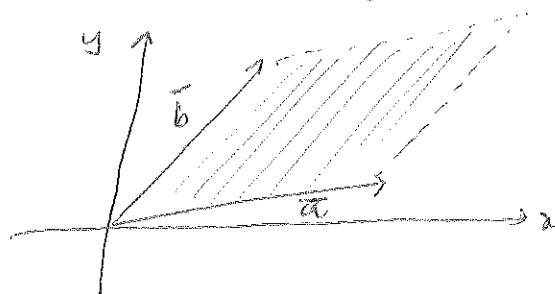
$$= \vec{i} \begin{vmatrix} 2 & 0 \\ -1 & 2 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix}$$

$$= \vec{i}(4 - 0) - \vec{j}(2 - 0) + \vec{k}(-1 - 6)$$

$$= 4\vec{i} - 2\vec{j} - 7\vec{k}$$

(4)

Note: $|\vec{a} \times \vec{b}|$ is the area of the parallelogram determined by non-zero vectors $\vec{a}$ and $\vec{b}$



Example: let $\vec{a} = \langle 4, 6, -2 \rangle$ and $\vec{b} = \langle 8, 12, -4 \rangle$

Find $\vec{a} \times \vec{b}$

What can you conclude

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & 6 & -2 \\ 8 & 12 & -4 \end{vmatrix} = \vec{i} \begin{vmatrix} 6 & -2 \\ 12 & -4 \end{vmatrix} - \vec{j} \begin{vmatrix} 4 & -2 \\ 8 & -4 \end{vmatrix} + \vec{k} \begin{vmatrix} 4 & 6 \\ 8 & 12 \end{vmatrix} \\ &= \vec{i}(-24 + 24) - \vec{j}(-16 + 16) + \vec{k}(48 - 48) \\ &= \vec{0} \end{aligned}$$

$$\Rightarrow \vec{a} \parallel \vec{b} \quad (\vec{a} \text{ parallel to } \vec{b})$$

We could have noticed since $\vec{b} = \langle 8, 12, -4 \rangle = \langle 4, 6, -2 \rangle$

$$\Rightarrow \vec{b} = 2\vec{a}$$

Example:

①

Find a vector of length 2 that is perpendicular to both

$$\vec{a} = \langle 1, -1, 2 \rangle \quad \text{and} \quad \vec{b} = \langle 3, 2, -1 \rangle$$

Solution: A cross product of two vectors in $\mathbb{R}^3$ is always a vector that is $\perp$ to both vectors.

ie if we can calculate $\vec{a} \times \vec{b}$

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 2 \\ 3 & 2 & -1 \end{vmatrix} = \vec{i} \begin{vmatrix} -1 & 2 \\ 2 & -1 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} \\ &= \vec{i}(1-4) - \vec{j}(-1-6) + \vec{k}(2+3) \\ &= -3\vec{i} + 7\vec{j} + 5\vec{k} \end{aligned}$$

• This vector $\langle -3, 7, 5 \rangle$ is $\perp$ to $\vec{a}$ and
also $\perp$ to $\vec{b}$

• To find a vector of length 2 we first find a unit vector

$$\vec{u} = \frac{1}{\sqrt{(-3)^2 + 7^2 + 5^2}} \langle -3, 7, 5 \rangle$$

Now a vector of length 2 is

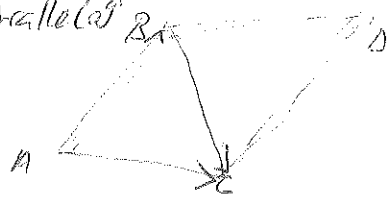
$$2\vec{u} = \left\langle \frac{-6}{\sqrt{83}}, \frac{14}{\sqrt{83}}, \frac{5}{\sqrt{83}} \right\rangle$$

(6)

Example: Find the area of the triangle with vertices

$$A(2, 3, 4), B(1, 2, 3) \text{ and } C(0, -2, -1)$$

Solution: Area of a $\Delta = \frac{1}{2}$ area of the Parallelogram
Find vectors $\vec{AB}$ and $\vec{AC}$.



Then Area of the triangle $= \frac{1}{2} |\vec{AB} \times \vec{AC}|$

$\left(\frac{1}{2} \cdot \text{length of vector } \vec{AB} \times \vec{AC} \right)$.

Properties of cross product. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be vectors in $\mathbb{R}^3$ and $c \in \mathbb{R}$ a scalar

1. $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

2. $(c\vec{a}) \times \vec{b} = c(\vec{a} \times \vec{b})$

3. $\vec{a} \times (c\vec{b}) = c(\vec{a} \times \vec{b})$

4. $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$

5. $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$

Recall
 $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

Example: Find a vector perpendicular to the plane that passes through the points
 $P(1, -1, 1), Q(2, 0, 1)$ and $R(2, -1, 1)$

Solution: A vector perpendicular to the plane
passing through the points P , Q and R is
vector $\overrightarrow{QP} \times \overrightarrow{QR}$

